

Transaction Analysis Script

Plain-English Code Walkthrough & Results Summary

This document explains, in simple English, what the transaction analysis Python script does step by step — and includes the actual results produced when it was run on your data.

1. Setup

- Imports the **pandas** library, used to work with tables of data.
- Reads the **transaction.csv** file into a table called a DataFrame (**df**).
- Removes any extra spaces from column names to avoid errors later on.

2. Converting the Date Column

- Converts the **Date** column into a proper date format so Python reads it as dates, not plain text.
- Any invalid dates are replaced with **NaT** (Not a Time) instead of crashing the program.
- Creates a new **Month** column from the Date column, so transactions can be grouped by month.

3. Calculating Total Spending

- Adds up every value in the **Amount** column.
- Displays the total amount spent.

Total Spending

Rs. 28,350.00

4. Calculating Monthly Totals

- Groups all transactions by the **Month** column.
- Calculates and displays the total spending for each month.

Result: Monthly Spending Breakdown

Month	Total Spent (Rs.)
2026-06	28,350

Highest spending month: June 2026 (Rs. 28,350) — currently the only month of data available. Once more months are added, this table will show which month had the highest and lowest spending.

5. Spending by Category

- Groups transactions by the **Category** column (e.g. Food, Transport, Shopping, Utilities).

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- Calculates the total amount spent in each category.
- Sorts categories from highest to lowest spending.
- Displays the category with the highest total spending.

Result: Category-wise Spending (Highest to Lowest)

Category	Amount (Rs.)
Groceries	7,000
Food	6,250
Utilities	6,200
Shopping	4,500
Entertainment	3,100
Transport	1,300

Biggest category: Groceries (Rs. 7,000.00)

6. Finding Recurring Payments

- Groups transactions by **Description** (e.g. Netflix, Spotify, Grocery Store).
- Counts how many times each description appears.
- Calculates the total amount spent on each description.
- Shows only descriptions appearing **more than once** — these are likely recurring payments or subscriptions.
- If none repeat, the script reports that no recurring payments were found.

Result: Recurring Payments Detected

Description	Times Paid	Total (Rs.)	Amount Each (Rs.)
Grocery Store	3	7,000	2,500
Foodpanda	2	3,600	1,800
Netflix	2	2,400	1,200
Uber	2	1,200	600
Spotify	2	700	350

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Grocery Store is your most frequent recurring expense, paid 3 times for a combined Rs. 7,000. Netflix, Spotify, Foodpanda, and Uber each recurred twice, together adding up to regular subscription and delivery costs.

7. Finding Duplicate Transactions

- Checks for transactions with the **same Description** and the **same Amount** appearing more than once.
- These could indicate repeated or accidental duplicate payments.
- If found, they are displayed sorted by description and amount.
- If none are found, the script prints "No duplicate transactions found."

Duplicate Transactions Found:

None — No duplicate transactions found.

No exact duplicate transactions (same description and same amount repeated) were found in this dataset. This means there's no evidence of an accidental double payment.

8. Final Message

- Displays "**Analysis Complete!**" once the script has finished running successfully.

Overall Summary

This script analyzes a transaction history stored in a CSV file. For the current dataset (June 2026), it found total spending of **Rs. 28,350**, with **Groceries** as the biggest spending category (Rs. 7,000). The most frequent recurring expense was **Grocery Store** (3 payments, Rs. 7,000 total), followed by Netflix, Spotify, Foodpanda, and Uber, each recurring twice. **No duplicate transactions** were detected, meaning there is currently no sign of accidental repeated payments.